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Effects of trace metal supplement form on production performance, antioxidant status, and mammary tissues during repeated intramammary challenges

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ABSTRACT

Mastitis negatively alters milk composition, partly due to oxidative damage to the mammary secretory epithelium. Trace metals are important cofactors involved in oxidative balance and immune function. However, trace metal absorption is affected by its supplement form. The objective of this study was to evaluate milk composition and indicators of oxidative balance in cows supplemented with sulfate or methionine hydroxy analog chelates of Zn, Cu, and Mn at isometal levels during sterile mastitis. A total of 36 mid-lactation multiparous Holstein cows were used. Cows received either (1) sulfate-bound trace metals and intramammary infusions of saline (SULF-CON; n = 9), (2) sulfate-bound trace metals and intramammary infusions of formalin-fixed *Staphylococcus aureus* (SULF-CHAL; n = 12), or (3) methionine hydroxy analog chelates of Zn, Cu, and Mn and intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL; n = 12). The study lasted 6 wk, and cows were repeatedly intramammarily infused every 3 d during the final 2 wk to elicit chronic mastitis. Milk yields and components were measured at each milking (twice daily) throughout the challenge period. Blood samples were collected during the final challenge, and mammary tissue biopsies were performed at the end of the study. Milk yields and DMI were similar between treatments. Milk SCS was markedly increased in the MTX-CHAL and SULF-CHAL cows in response to challenge. Milk lactose content decreased in MTX-CHAL and SULF-CHAL cows but tended to be greater in MTX-CHAL cows. The plasma ferric reducing antioxidant power values of SULF-CHAL cows were lower than SULF-CON, whereas the values of MTX-CHAL cows were intermediate. Histological

evaluation of mammary tissue structure and markers of lipid peroxidation did not differ between treatments. Results indicate that repeated infusions of formalin-fixed *Staph. aureus* negatively affected milk composition, and that trace mineral supplement form may better maintain antioxidant status and some markers of epithelial barrier competency during mastitis.

Key words: mastitis, minerals, antioxidant status, chelates

INTRODUCTION

Mastitis, inflammation of the mammary gland, remains one of the most prevalent and costly diseases in the dairy industry. Mastitis results in financial losses through decreased milk yield and unfavorable changes in milk composition from affected quarters (Schultz, 1977; Blosser, 1979; Kitchen, 1981). The duration of a mastitis event can be short or prolonged, with prolonged being considered chronic mastitis.

Chronic mastitis exacerbates production losses compared with nonchronic cases (Martins et al., 2020), partly due to cumulative damage to mammary secretory epithelium (Gudding et al., 1984). During the early inflammatory response, neutrophils infiltrate the mammary gland and generate and release reactive oxygen species (ROS; Miller et al., 1993). Endogenous antioxidant enzymes can convert ROS into less reactive metabolites as a means to stave off ROS-induced tissue damage. Sustained or excessive ROS production, however, can exceed the cow's antioxidant capacity, leading to oxidative imbalance (Sordillo and Aitken, 2009). Such an imbalance can result in damaged mammary epithelium (Sordillo and Aitken, 2009), which may progressively lead to replacement of secretory tissue with fibrotic tissue and reduce the secretory capacity of the mammary glands (Chandler and Reid, 1973; Heald, 1979; Nickerson and Heald, 1981).

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Trace metals such as zinc (Zn), copper (Cu), and manganese (Mn) are essential for supporting immune function, maintaining epithelial competency, and restoring oxidative balance (Miller et al., 1993; Spears and Weiss, 2008). Trace metals have traditionally been supplemented as inorganic mineral salts, such as sulfates. However, these metals can readily dissociate in the rumen from their mineral complex, increasing their likelihood of interacting with dietary antagonists and forming insoluble, poorly absorbed metal complexes (Spears, 2003). Organic-bound trace metals, in which the metal ion is bound to an organic ligand, have received increasing attention as a more bioavailable alternative to inorganic mineral supplementation. Feeding Zn, Cu, and Mn bound to organic moieties can result in greater intestinal absorption of the metal ion (Predieri et al., 2005; Wright et al., 2008), allowing improved trace metal status at lower or equivalent dietary inclusion rates (Nemec et al., 2012). Supplementation of chelated forms of Zn, Cu, and Mn has been associated with improved production performance, including increased milk yield (Kellogg et al., 2003, 2004; Nocek et al., 2006) and reduced SCC (Kellogg et al., 2003, 2004; Kinal et al., 2007), as well as enhanced antioxidant capacity via increased superoxide dismutase activity (Zhao et al., 2015). In particular, metal methionine hydroxy analog chelates, a form of organic-bound trace metals, have demonstrated improved reproduction and reductions in lameness (Bach et al., 2015). Conversely, other studies have reported no significant benefits when feeding trace metal chelates instead of sulfates (Nemec et al., 2012; Yasui et al., 2019).

The conflicting results may arise from differences in study duration and the physiological status of the cows used (Rabiee et al., 2010). Gaining a better understanding of the effects of feeding organic-bound trace metals during a sustained immune challenge is necessary to determine if organic-bound trace metals improve chronic disease resolution relative to feeding sulfate-bound trace metals. Our first objective was to determine the effect of repeated formalin-fixed *Staphylococcus aureus* intramammary infusions on production performance, markers of blood and tissue oxidative balance, and mammary tissue structure. Our second objective was to determine if feeding organic-bound trace metals reduces the magnitude of negative outcomes resulting from repeated infusions of formalin-fixed *Staph. aureus*. We hypothesized that cows challenged with formalin-fixed *Staph. aureus* would exhibit reduced milk yield and impaired oxidative balance, and that supplementing methionine hydroxy analog chelates of Zn, Cu, and Mn would reduce the magnitude of milk composition changes and improve antioxidant balance during chronic mastitis relative to cows fed sulfate forms of Zn, Cu, and Mn.

MATERIALS AND METHODS

Animal Selection and Study Design

A total of 36 multiparous Holstein cows (60–140 DIM) were used. The study commenced in January 2024 and concluded in October 2024. To be eligible for enrollment, cows must have been clinically healthy and had an SCC <200,000 cells/mL. Cows were housed in individual tie-stalls at the Ohio Agriculture Research and Development Krauss Dairy facility (Wooster, OH) and allowed a 7-d acclimation period before the start of the study. Cows were blocked by DIM and milk yield into 12 blocks, each containing 3 cows, where they were randomly assigned to 1 of 3 treatments and subjected to the 6-wk experimental timeline shown in Figure 1. Cows began the study consuming the same basal diet containing sulfate-bound trace metals, which was subsequently top dressed with a trace metal absorptive antagonist (experimental d –7 to –1) before beginning to receive their experimental diets for the study duration (experimental d 0–41). During the last 2 wk of the study (experimental d 28–41), cows were repeatedly administered intramammary infusions of either saline or formalin-fixed *Staph. aureus* every 3 d. The 3 unique diet-intramammary infusion treatment combinations were (1) control cows fed a diet containing sulfate-bound trace metals that received repeated intramammary infusions of saline (SULF-CON; n = 12); (2) cows fed the same diet containing sulfate-bound trace metals that were repeatedly challenged with intramammary infusions of formalin-fixed *Staph. aureus* (SULF-CHAL; n = 12); and (3) cows fed a diet containing methionine hydroxy analog chelates of Zn, Cu, and Mn (Mintrex, Novus International Inc., Chesterfield, MO) that received repeated intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL; n = 12). The decision to use 12 cows per treatment group was based on the number of cows used by Yasui et al. (2019), and no new power analysis was conducted. Cows were milked twice per day and fed once per day, with ad libitum access to water and feed throughout the study. Body weights were measured on d –7, 0, 28, and 41.

Experimental Diets and Proximate Analysis

Nutrient and chemical compositions of the diets fed are shown in Table 1. All cows were fed the sulfate-bound trace metal diet for 1 wk (d –7 to –1), which was top dressed with additional sulfur (increasing diet to 0.4% sulfur on a DM basis), 300 ppm of iron from iron sulfate, and 5.0 ppm molybdenum from sodium molybdate to reduce trace mineral absorption and status, as in Yasui et al. (2019). On d 0, top dressing was discontinued, reverting to a sulfur inclusion of 0.25% (DM basis), and

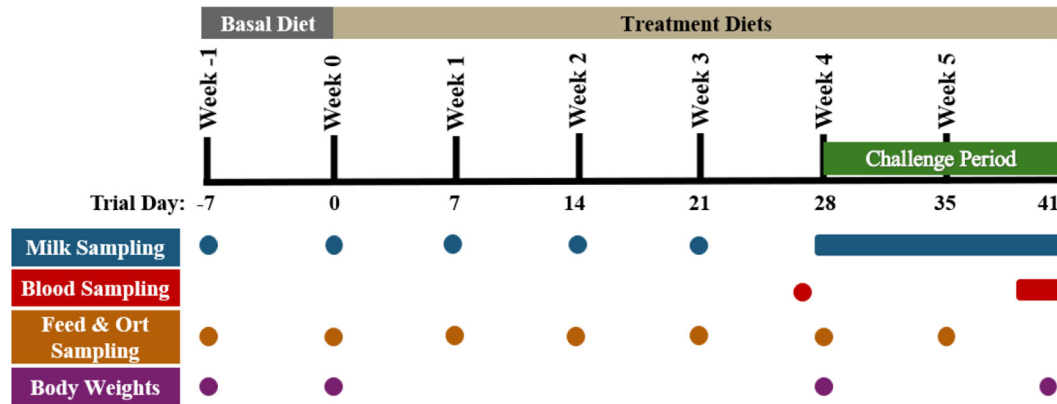


Figure 1. Experimental timeline for the 36 cows used in the study. The timeline illustrates the periods during which the basal and treatment diets were administered, the start of the intramammary challenge period, and the collection points for milk, blood, and feed samples. Body weights were also recorded at the displayed time points. Each black line represents the trial week and corresponding trial day.

the MTX-CHAL cows began receiving the methionine hydroxy analog chelates containing diet. Treatment diets were then fed for 5 wk (experimental d 0 to 41). Both diets were offered as a TMR and consisted of corn silage (19% of diet DM), alfalfa silage (33% of diet DM), grass hay (2% of diet DM), and a concentrate mixture (46% of diet DM). Energy and trace metal availability requirements were calculated, and diets were subsequently formulated to meet or exceed the cow's nutritional requirements using the NASEM (2021) recommendations for maintenance and production; calculations were based on an average cow BW of 750 kg, consuming 28 kg of DMI, and producing 49.9 kg of milk per day. The diet containing sulfate-bound Zn, Cu, and Mn, was formulated to provide 339.5, 12.9, and 3.4 mg of available Zn, Cu, and Mn per day, respectively. The chelate containing diet fed to MTX-CHAL cows was formulated to be iso-metal and iso-methionine relative to the sulfate-bound metal diet. Feed orts were weighed and recorded daily. Concentrate, forage, and ort samples were collected weekly, composited across the study, and stored at -20°C . Dry matter content was determined by drying samples at 55°C , and used to calculate daily DMI. Composited samples of forages and concentrates were submitted to Rock River Laboratory (Watertown, WI) for wet chemistry analysis.

Intramammary Challenge

We chose to use formalin-fixed *Staph. aureus* as our challenge agent because our goal was to induce mild clinical mastitis that could be maintained for an extended time period via repeated infusions. The growth and fixation of the *Staph. aureus* followed the procedures detailed by us previously (Gammariello et al., 2024). Briefly, stocks of formalin-fixed *Staph. aureus* were prepared and stored at -80°C until needed. The night before

Table 1. Ingredient and chemical composition of experimental diets fed to cows

Item	Sulfate diet	MTX diet
Ingredient, % of diet ¹		
Alfalfa silage	19.1	19.1
Corn silage	33.2	33.2
Grass hay	2.0	2.0
Corn, ground	18.5	18.6
Amino Plus ²	6.54	6.49
Soybean meal	9.28	9.46
Soybean hulls	6.26	6.26
Megalac ³	2.03	2.03
Rumensin 90 ⁴	0.007	0.007
Methionine ⁵	0.024	—
Vitamin and mineral mix ⁶	2.95	2.59
Cu sulfate	0.0009	—
Mn sulfate	0.0005	—
Zn sulfate	0.0091	—
Mintrex Cu ⁷	—	0.0018
Mintrex Mn ⁷	—	0.0005
Mintrex Zn ⁷	—	0.0205
Chemical composition		
DM, %	50.6	50.5
CP, %	18.6	18.4
NDF, %	30.6	29.8
ADF, %	19.4	19.2
Starch, %	28.6	29.5
Ash, %	9.6	9.0
Cu, ppm	9.6	11.4
Mn, ppm	52.9	49.6
Zn, ppm	89.3	76.2

¹All values are based on a DM percentage.

²High protein soybean meal (Ag Processing Inc., Omaha, NE).

³Calcium salts of palm oil (Volac Wilmar Feed Ingredients, Orwell, GB).

⁴Monensin at 41.13 g per kg of product (Elanco Animal Health, Indianapolis, IN).

⁵Methionine = HMTBa methionine source (MFP; Novus International).

⁶Vitamin and mineral mix contained 7.7% trace mineral salts, 40.4% sodium bicarbonate, 12.3% limestone, 4.2% selenium, 7.9% dicalcium phosphate, 3.3% dynamite, 7.7% biotin, 3.9% sodium bisulfate, 12.4% magnesium oxide (sulfate diet only) and <1% vitamin E, vitamin A, vitamin D.

⁷Cu, Mn, and Zn methionine hydroxy analog chelates (Novus International).

intramammary infusion, stock tubes of formalin-fixed *Staph. aureus* were defrosted at 5°C and used to prepare syringes that contained 2 billion cfu of formalin-fixed *Staph. aureus* suspended in 20 mL of sterile PBS (cat # SH30256.01, Cytiva, Marlborough, MA). Syringes containing only sterile PBS were also prepared for control cows. The rear quarters of all cows were infused with their respective treatments beginning d 28 after morning milking; infusions were repeated every 3 d thereafter, for a total of 5 infusions. Infusions followed the procedures detailed in Enger et al. (2018).

Milk Production and Composition

Composite whole milk samples were collected weekly from d 0 to 27 at both the morning and evening milkings, and then every milking during the intramammary challenge period (i.e., d 28 to 41). Milk samples were collected into DHIA-provided tubes containing preservative pellets (44% Bronopol, 2% Natamycin) and stored at 5°C until analysis. Milk samples were analyzed for SCC and milk components by a local DHIA laboratory (Columbus, Ohio) using a Bentley Fourier-transform infrared spectrometer and flow cytometer.

Blood Sampling and Analysis

Jugular blood was collected the day preceding intramammary infusions to serve as a prechallenge baseline on d 27, and then again on d 40 and 41 at 0, 3, 6, 9, 12, 18, 24, 30, and 36 h after the final infusion. For blocks 1 to 3, the final challenge blood samples were obtained by jugular venipuncture, but jugular catheters were used for all subsequent blocks (blocks 4–12). Jugular catheters (catalog no. J1042, Jorgensen Laboratories Inc., Loveland, CO) were placed by a veterinarian using a local anesthetic (0.5–1.0 mL of 2% lidocaine at the insertion site). Before sampling, catheter lines were flushed by drawing and discarding ~10 mL of blood. Blood was then collected into 10-mL EDTA tubes, inverted at least 5 times, and placed on ice. Catheters were flushed with ~10 mL of heparinized saline (20 IU/mL) after each collection. Blood tubes were centrifuged at $1,600 \times g$ for 15 min at 4°C to obtain plasma, which was aliquoted into 1.5-mL microcentrifuge tubes and stored at –20°C until analysis. Plasma antioxidant capacity was determined using a ferric reducing antioxidant power (FRAP) assay described by Gammariello et al. (2024), which was adapted from Shangraw et al. (2020). Plasma was diluted 1:1 with distilled water and combined with FRAP reagent (0.3 M acetate buffer, 10 mM 2,4,6-tripyridyl-s-triazine, and iron trichloride) in triplicate in a 96-well plate. Plates were incubated at 37°C and absorbance values were measured after 15 min at 593 nm. The intra-assay and

interassay coefficients of variation were 3.6% and 1.3%, respectively.

Mammary Tissue Collection and Analysis

Mammary tissue samples were obtained after the last milking on d 41 from either a left or right rear quarter from all cows in blocks 2, 4, and 10, yielding a total of 9 cows biopsied (i.e., 3 per treatment group). Biopsies were performed using a tool and technique similar to Farr et al. (1996) as we detailed previously (Gammariello et al., 2024). A portion of the collected tissues was fixed in 10% neutral buffered formalin for 48 h, and the remaining tissues were snap-frozen in liquid nitrogen and stored at –80°C. Formalin-fixed tissues were paraffin-embedded for histological evaluation via hematoxylin and eosin staining as in Gammariello et al. (2024). Snap-frozen mammary tissue samples were analyzed for oxidative status using a commercial thiobarbituric acid reactive substances (TBARS) assay (catalog no. 10009055, Cayman Chemical, Ann Arbor, MI) following the manufacturer's instructions. Tissues were homogenized using a Biomasher (catalog no. 1215D63, Kimble, Millville, NJ) in radioimmunoprecipitation assay buffer (catalog no. 10010263, Cayman Chemical), then mixed with sodium dodecyl sulfate and the supplied color reagent. Samples were analyzed in duplicate in a black 96-well plate, and absorbance was measured at 530 nm. The intra-assay coefficient of variation was 3.9%. Results were expressed as malondialdehyde (MDA) concentrations.

Statistical Analysis

All response variables were analyzed using PROC GLIMMIX in SAS 9.4 (SAS Institute Inc.). Milk SCC were transformed into SCS as described by Norman et al. (2000); SCC <12,500 cells/mL were increased to this minimum value because lesser values cannot be transformed. The fixed explanatory effects were treatment, time, and their interaction, and the random effects were block and cow nested within block. The denominator degrees of freedom adjustment described by Kenward and Roger (1997) was used for all models. Measures were repeated on cows over time; the following covariance structures were tested for each response variable: compound symmetry, heterogeneous compound symmetry, first order autoregressive, heterogeneous first order autoregressive, and Toeplitz. When time point intervals were uneven, only variance components, compound symmetry, and heterogeneous compound symmetry covariance structures were tested. Covariance structures were independently selected for each response variable based on the lowest corrected Akaike information criterion values (Hurvich and Tsai, 1989). Least squares means

resulting from the models were contrasted using Fisher's least significant difference test. Differences were considered significantly different at $P \leq 0.05$ and marginally different at $0.05 < P \leq 0.15$.

RESULTS AND DISCUSSION

Response to Challenge

Three SULF-CON cows were removed from the study due to developing mastitis, resulting in 9 SULF-CON cows, 12 SULF-CHAL cows, and 12 MTX-CHAL cows that completed the study and whose data were analyzed.

All cows that received intramammary infusions of formalin-fixed *Staph. aureus* developed clinical signs of mastitis at the second or subsequent challenges. Abnormal milk (i.e., flakes, clots, or chunks) was observed in all SULF-CHAL and MTX-CHAL cows within 12 to 24 h after the second or subsequent infusion. Redness and swelling of affected quarters were noted in some challenged cows, whereas a febrile response infrequently occurred.

Dry Matter Intake and Production Responses

Body weights ($P = 0.83$, data not shown) and DMI ($P = 0.58$) did not differ between treatment groups. Mean DMI was 30.78 ± 1.73 kg for MTX-CHAL, 29.59 ± 1.74 kg for SULF-CHAL, and 32.37 ± 2.00 kg for SULF-CON cows.

Milk SCS values are presented in Figure 2. Milk SCS did not differ between treatment groups before the first challenge on d 28 ($P = 0.70$) but were affected by intramammary infusion treatment and intramammary treatment interacting with time during the challenge period ($P < 0.01$). Milk SCS were similar among treatment groups at the time of challenge ($P = 0.80$). After the first intramammary challenge, milk SCS of MTX-CHAL and SULF-CHAL cows rose and remained greater than that of SULF-CON cows for the remainder of the trial ($P < 0.01$). The mean SCS of MTX-CHAL and SULF-CHAL did not differ (6.06 vs. 5.97 ± 0.15 ; $P = 0.67$). The greater SCS of both MTX-CHAL and SULF-CHAL cows relative to SULF-CON throughout the challenge aligns with our previous observations (Enger et al., 2025) and indicates repeated challenges of formalin-fixed *Staph. aureus* initiated a sustained immune response. The lack of differences in SCS between MTX-CHAL and SULF-CHAL cows indicates that the form of trace metals did not appreciably influence somatic cell response to the repeated intramammary challenges. Similar to in this study, other studies have reported no effect of organic-bound metal supplementation on SCC in response to sterile mastitis (Yasui et al., 2019).

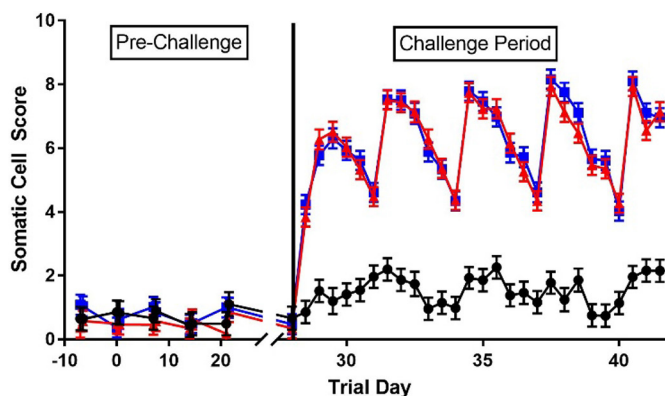


Figure 2. Mean SCS of cows receiving either sulfate-bound trace metals and repeated intramammary infusions of saline (SULF-CON; black circles, $n = 9$), sulfate-bound trace metals and repeated intramammary infusions of formalin-fixed *Staph. aureus* (SULF-CHAL; red triangles, $n = 12$), or chelated trace metals and repeated intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL; blue squares, $n = 12$) before the challenge period (d -7 to d 27) and during the challenge period (d 28 to d 41). Error bars denote SEM.

Milk yields are presented in Figure 3A. Milk yields did not differ between treatment groups before ($P = 0.82$) or during ($P = 0.68$) the challenge period. Mean milk yields before the challenge period were 25.4 ± 0.8 kg for MTX-CHAL, 24.9 ± 0.8 kg for SULF-CHAL, and 25.3 ± 0.9 kg for SULF-CON cows and 23.8 ± 0.7 kg for MTX-CHAL, 23.7 ± 0.7 kg for SULF-CHAL, and 24.5 ± 0.8 kg for SULF-CON cows during the challenge period. The outcome that milk yields of SULF-CON cows were not greater than SULF-CHAL and MTX-CHAL cows during the challenge period was unexpected. Using a split-udder model, we previously reported that repeatedly infusing udder halves with the same regimen and dosage of formalin-fixed *Staph. aureus* reduced milk yield by an average of 0.5 kg/milking relative to saline-infused udder halves, culminating to a 1.0 kg loss per milking after the final intramammary challenge (Enger et al., 2025). The between-cow variability of the present study (i.e., not having a split-udder model where each cow serves as its own control) likely reduced the ability to detect milk yield differences. A greater number of cows, stronger inflammatory response, or greater duration of intramammary challenge may have been required to detect significant milk yield losses when a split-udder model is not used. Although no differences in milk yields were observed between the SULF-CHAL and MTX-CHAL groups, previous studies have reported greater milk production in cows supplemented with MTX or other organic-bound trace metals under various conditions. A meta-analysis of 20 studies reported increased milk yield in response to feeding trace metals bound to organic moieties (Rabiee et al., 2010), but high variability across studies existed,

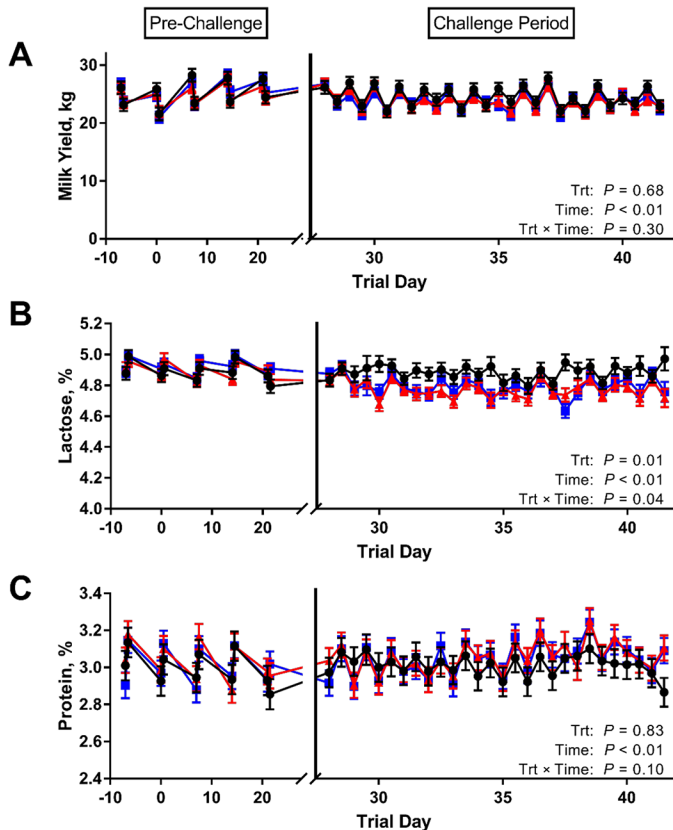


Figure 3. Mean milk yield (A), lactose content (B), and protein content (C) of whole milk from cows receiving either sulfate-bound trace metals and repeated intramammary infusions of saline (SULF-CON; black circles, $n = 9$), sulfate-bound trace metals and repeated intramammary infusions of formalin-fixed *Staph. aureus* (SULF-CHAL; red triangles, $n = 12$), or chelated trace metals and repeated intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL; blue squares, $n = 12$) before the challenge period (d -7 to d 27) and during the challenge period (d 28 to d 41). The P -values presented in each panel correspond to the effects of the dietary treatment and intramammary infusion (Trt), trial d (Time), and their interaction (Trt $\times$ Time) during the challenge period (d 28 to d 41); P -values are not presented for the data preceding the challenge period (d -7 to d 27) because no meaningful Trt, or Trt $\times$ Time, effect was detected. Error bars denote SEM.

which likely resulted from differences in organic moiety structure and supplementation duration.

Milk lactose content is shown in Figure 3B. Before the challenge period, milk lactose content did not differ between treatments ($P = 0.41$). During the challenge period, milk lactose content was lesser in MTX-CHAL ($P = 0.01$; $4.79\% \pm 0.03\%$) and SULF-CHAL ($P < 0.01$; $4.77\% \pm 0.03\%$) than SULF-CON cows ($4.89\% \pm 0.03\%$); MTX-CHAL and SULF-CHAL cows had similar lactose content ($P = 0.54$) when the entire 2-wk challenge period is considered. However, following the final intramammary infusion, SULF-CON cows had a greater lactose content than SULF-CHAL cows ($P < 0.01$), whereas the lactose content of MTX-CHAL cows

($4.83\% \pm 0.03\%$) was intermediate, and marginally less than SULF-CON ($4.91\% \pm 0.04\%$; $P = 0.08$) and marginally greater than SULF-CHAL ($4.76\% \pm 0.032\%$; $P = 0.08$). The reduction in milk lactose content observed in intramammarily challenged cows was expected. We and others have previously reported decreases in milk lactose content in response to intramammary challenges (Akers and Thompson, 1987; Gammariello et al., 2024) and naturally occurring mastitis (Schultz, 1977). Reduced milk lactose concentrations can result from a compromised epithelial barrier. Indeed, milk lactose concentration is commonly used as an indicator for epithelial barrier competency because milk lactose can diffuse out of the alveolar lumen into the interstitial space, reducing its concentration in milk (Wellnitz and Bruckmaier, 2021). Cows fed MTX-bound trace metals tended to have higher lactose concentrations following the final challenge compared with those fed sulfates, suggesting that MTX trace metals may help better maintain epithelial barrier competency during a chronic mastitis event.

Milk protein content is presented in Figure 3C. Milk protein content did not differ between treatment groups before the challenge period ($P = 0.83$) or during the challenge period ($P = 0.82$), but we observed a marginal effect for a treatment by time interaction ($P = 0.10$), with challenged cows beginning to have slightly greater milk protein content than controls during the final challenge. We have previously discerned differences in milk protein content in response to repeated infusions of formalin-fixed *Staph. aureus* when using a split-udder half model (Enger et al., 2025). Variability between cows likely interfered with our ability to detect significant differences in milk protein in this study. A shift in protein composition occurs during mastitis: concentrations of mammary-synthesized proteins, such as casein, tend to decrease, whereas concentrations of serum-derived proteins, including serum albumin and immunoglobulins, increase (Kitchen, 1981). Depending on the relative magnitude of these protein fraction shifts, total milk protein content often remains stable or may slightly increase during mastitis (Schultz, 1977). Milk protein content also did not differ between dietary treatment, which is consistent with the reporting of Rabiee et al. (2010), who conducted a meta-analysis and found no significant differences in milk protein content between cows supplemented with trace metals bound to organic versus inorganic compounds, noting substantial variability among study results.

Antioxidant Status

Plasma FRAP concentrations were quantified to assess the oxidative balance of cows repeatedly challenged; FRAP responses are presented in Figure 4. Plasma FRAP values did not differ among treatment groups before the challenge period ($P = 0.59$), but marginally differed during the final intramammary challenge ($P = 0.13$). Mean plasma FRAP values during the final challenge period were 252 ± 6.7 for MTX-CHAL, 245 ± 6.7 for SULF-CHAL, and 263 ± 7.5 for SULF-CON cows. Plasma FRAP values of MTX-CHAL cows did not differ from SULF-CHAL ($P = 0.38$) or SULF-CON cows ($P = 0.20$), but SULF-CHAL cows had lower plasma FRAP values than SULF-CON cows ($P = 0.05$). The lack of significant difference in FRAP concentrations between MTX-CHAL and SULF-CHAL cows is consistent with findings from Yasui et al. (2019), who also reported similar total antioxidant capacity between cows fed either organic- or inorganic-bound trace metals during a sterile immune challenge. Complicating this observation, however, is that SULF-CHAL cows had significantly lower FRAP values than SULF-CON cows, but MTX-CHAL and SULF-CON cows had similar FRAP values. Taken together, this suggests an incomplete picture in which the MTX-CHAL cows were better able to maintain their antioxidant status than SULF-CHAL cows. Further research is needed to clarify the effects of MTX trace metal supplementation on antioxidant status. Individual cow variability may

have limited the ability to detect significant differences between dietary treatments; a larger sample size may be needed to conclude treatment effects.

Mammary Tissue Responses

Mammary tissue TBARS responses are presented in Figure 5A. No differences were observed among treatment groups ($P = 0.38$). This outcome was unexpected, as previous work by Eslami et al. (2015) found that LPS infusions into rat mammary glands resulted in significantly elevated mammary MDA concentrations. The milder immunostimulant (formalin-fixed *Staph. aureus*) used here may have been insufficient to induce measurable oxidative damage in mammary tissue. Additionally, the TBARS assay, although commonly used to assess lipid peroxidation in ruminants, lacks specificity for MDA and can react with other metabolites (Celi, 2011). This lack of specificity may have also contributed to the absence of detectable differences between treatments in this study.

The collected mammary tissues were qualitatively assessed via hematoxylin and eosin staining and did not present any obvious differences in tissue structure. Mammary tissues from MTX-CHAL and SULF-CHAL appeared similar, with both presenting lobules that would appear normal and highly secretory and then separate lobules exhibiting leukocyte infiltration and clotted milk. Abundance of luminal space, epithelial area, and stromal tissue area did not noticeably differ between treatment groups. Such an outcome is not surprising given the absence of an appreciable reduction in milk yield.

General Discussion

Overall, the results of this study partially support our hypothesis that cows infused with formalin-fixed *Staph. aureus* would exhibit reduced milk yields and compromised oxidative status. Unexpectedly, repeated challenges of formalin-fixed *Staph. aureus* failed to appreciably affect milk yield or trigger lipid peroxidation in mammary tissue which is likely due to uncontrolled variability, chiefly between cow milk production. On the other hand, milk lactose content was significantly altered, indicating that the intramammary challenges did compromise milk synthesis and epithelial barrier competency. Additionally, we hypothesized that cows supplemented with MTX trace metals would show less alterations in milk composition and maintain better antioxidant status in response to the challenge compared with cows fed sulfate-bound trace metals. Cows receiving MTX trace metals exhibited smaller reductions in milk lactose concentrations and showed greater FRAP values than cows fed sulfate-bound trace metals, suggesting that MTX trace metal supplementation may better maintain epithe-

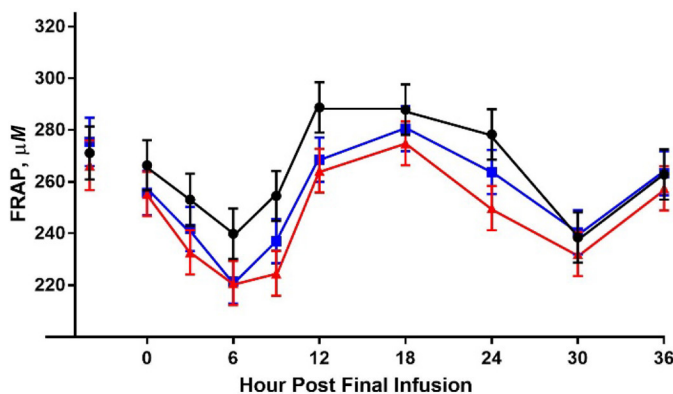


Figure 4. Mean ferric reducing antioxidant power (FRAP) values of cows fed either sulfate-bound trace metals and repeated intramammary infusions of saline (SULF-CON; black circles, $n = 9$), sulfate-bound trace metals and repeated intramammary infusions of formalin-fixed *Staph. aureus* (SULF-CHAL; red triangles, $n = 12$), or chelated trace metals and repeated intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL; blue square, $n = 12$). Mean FRAP values are for the final intramammary challenge; the values presented before h 0 are pre-challenge baseline values. The FRAP values did not differ among intramammary infusion and diet treatment groups preceding challenge ($P = 0.59$); P -values for the effects of treatment group, hour after intramammary infusion, and their interaction were $P = 0.13$, $P < 0.01$, $P = 0.72$, respectively. Error bars denote SEM.

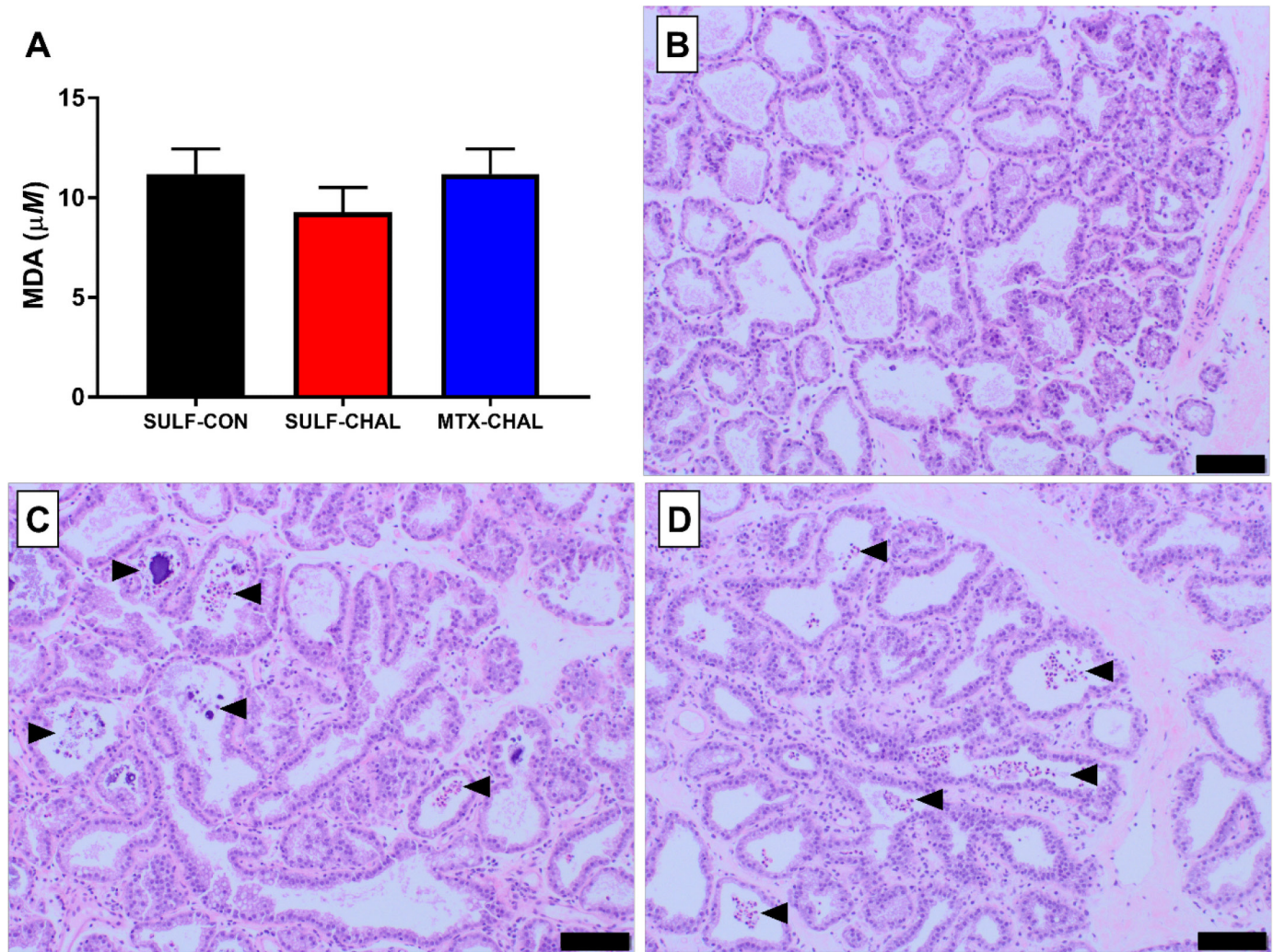


Figure 5. Mean malondialdehyde (MDA) concentrations of mammary tissues (A) of cows fed either sulfate-bound trace metals and intramammary infusions of saline (SULF-CON; black bar, $n = 3$), sulfate-bound trace metals and intramammary infusions of formalin-fixed *Staph. aureus* (SULF-CHAL; red bar, $n = 3$), or chelated trace metals and intramammary infusions of formalin-fixed *Staph. aureus* (MTX-CHAL; blue bar, $n = 3$). Mean MDA concentrations did not meaningfully differ between treatment groups ($P = 0.38$). Error bars denote SEM. Representative histological images are presented for mammary tissue collected 36 h after the final infusion from SULF-CON (B), SULF-CHAL (C), and MTX-CHAL (D) cows. Arrowheads identify areas of leukocyte infiltration and coagulated casein in C and D, which was commonly observed in mammary glands receiving intramammary infusions of formalin-fixed *Staph. aureus*. Scale bar = 100 μm .

lial barrier competency and antioxidant status. Notably, an interesting finding in this study is that cows infused with sterile saline still exhibited a measurable increase in SCS, suggesting that even saline infusions may provoke a mild immune response.

Limitations

Milk yield did not significantly differ between challenged and control cows in this study. This may be a result of greater than expected between-cow variability and the loss of 3 SULF-CON cows during the study. Additionally, the mild nature of the challenge may have limited

the ability to observe potential responses in epithelial barrier competency and tissue damage between treatment groups that may have resulted if a more severe disease state was induced. Increasing the frequency or duration of the formalin-fixed *Staph. aureus* infusions, or using or a live pathogen, might have elicited a stronger response; however, using a viable pathogen would likely result in highly variable durations of infection, complicating the central goal of this study. It is important to emphasize that this study provides no indication about how form of trace metals fed can affect immune cell competency and function (e.g., ROS production, phagocytosis, chemotaxis) as it relates to pathogen clearance and disease

resolution. Such a goal would require the use of a live pathogen.

CONCLUSIONS

This study evaluated the effects of repeated intramammary infusions of formalin-fixed *Staph. aureus* on milk production and oxidative status, in addition to mammary tissue structure, and whether these factors were influenced by form of trace metals fed. Repeated intramammary infusions of formalin-fixed *Staph. aureus* elicited marked SCS responses but failed to markedly reduce milk production. Challenged cows exhibited a reduction in milk lactose content; those supplemented with MTX-bound trace metals tended to maintain greater lactose content compared with cows fed sulfates. Limited evidence suggests that cows fed MTX-bound trace metals better maintained antioxidant status in response to the mastitis challenge. In conclusion, repeated intramammary infusions of formalin-fixed *Staph. aureus* did compromise some aspects of epithelial barrier competency, although a stronger immunostimulant or more frequent infusions may be necessary to elicit a measurable milk yield response. Supplementation with MTX trace metals may offer better support for maintaining epithelial barrier competency during mastitis, but further research is needed to fully clarify their ability to maintain antioxidant status during mastitis.

NOTES

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Nonstandard abbreviations used: FRAP = ferric reducing antioxidant power; MDA = malondialdehyde; ROS = reactive oxygen species; SULF-CHAL = sulfate-bound trace metals and intramammary infusions of formalin-fixed *Staphylococcus aureus*; SULF-CON = sulfate-bound trace metals and intramammary infusions of saline; SULF-MTX = methionine hydroxy analog chelates of Zn, Cu, and Mn and intramammary infusions of formalin-fixed *Staph. aureus*; TBARS = thiobarbituric acid reactive substances.

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